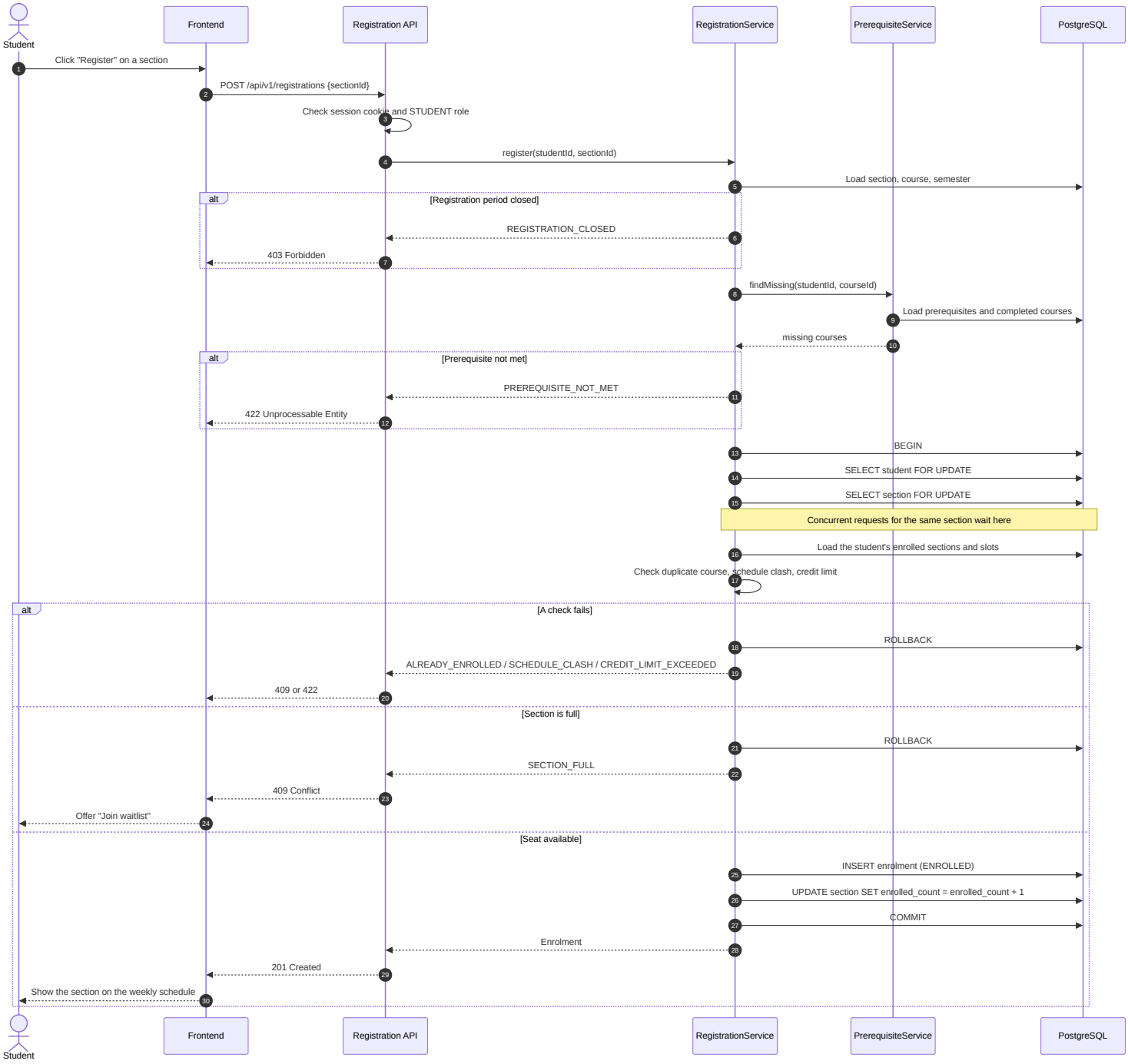

Student
Frontend
Registration API
RegistrationService
PrerequisiteService
PostgreSQL
1
Click "Register" on a section
2
POST /api/v1/registrations {sectionId}
3
Check session cookie and STUDENT role
4
register(studentId, sectionId)
5
Load section, course, semester
alt
[Registration period closed]
6
REGISTRATION_CLOSED
7
403 Forbidden
8
findMissing(studentId, courseId)
9
Load prerequisites and completed courses
10
missing courses
alt
[Prerequisite not met]
11
PREREQUISITE_NOT_MET
12
422 Unprocessable Entity
13
BEGIN
14
SELECT student FOR UPDATE
15
SELECT section FOR UPDATE
Concurrent requests for the same section wait here
16
Load the student's enrolled sections and slots
17
Check duplicate course, schedule clash, credit limit
alt
[A check fails]
18
ROLLBACK
19
ALREADY_ENROLLED / SCHEDULE_CLASH / CREDIT_LIMIT_EXCEEDED
20
409 or 422
[Section is full]
21
ROLLBACK
22
SECTION_FULL
23
409 Conflict
24
Offer "Join waitlist"
[Seat available]
25
INSERT enrolment (ENROLLED)
26
UPDATE section SET enrolled_count = enrolled_count + 1
27
COMMIT
28
Enrolment
29
201 Created
30
Show the section on the weekly schedule
Student
Frontend
Registration API
RegistrationService
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